

Gregory J. Gilbert

Postdoctoral Scholar

California Institute of Technology

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Curriculum Vitae last updated: 10 November 2025

Research Interests

exoplanets | planet formation | orbital dynamics | computational methods | science pedagogy

Appointments

Postdoctoral Scholar , California Institute of Technology	2025-present
Postdoctoral Scholar , University of California, Los Angeles	2021-2025
NASA FINESST Fellow , University of Chicago	2020-2021
Doctoral Candidate , University of Chicago	2016-2021

Education

PhD in Astronomy & Astrophysics , University of Chicago	2014-2021
BA in Physics , Washington University in St. Louis	2004-2008

Teaching

LSST Discovery Alliance Data Science Fellowship Program (Pittsburg, PA)	2025
Lecture: "Hamiltonian Monte Carlo, Nested Sampling, and Ensemble Samplers"	
Lecture: "Hierarchical Bayesian Analysis"	
Lecture: "Intractable Posteriors and Inference for Real World Problems"	

Teaching Assistant , University of Chicago (Chicago, IL)	2014-2020
ASTR: The Big Bang (2020) — ASTR: Galaxies (2017) — ASTR: Stars (2016)	
PHYS: Introduction to Astronomy (2015) — CHEM: Chemistry of Earth's Atmosphere (2014)	

Head of Science Curriculum , Academic Approach (Chicago, IL)	2011-2014
Led a team of six teachers and writers to produce lesson plans and exams covering physics, chemistry, biology, and earth sciences for use by over 100,000 Chicago Public Schools high school students. Recruited, vetted, and trained new team members.	

Science/Math Instructor , Academic Approach (Chicago, IL)	2008-2014
Instructed grade 9-12 students at all levels of math and science, including algebra, geometry, calculus, statistics, physics, and chemistry. Guided students on college applications.	

Advising

Sheila Sagar , University of Florida PhD	2024-2025
Paige Entrican , UCLA undergrad	2023-2025
Judah Van Zandt , UCLA PhD	2021-2025
Mason MacDougall , UCLA PhD	2021-2023
Devin Hoover , University of Chicago undergrad	2020-2021

Observing

Keck/KPF, 21 nights	2023-present
Keck/HIRES, 3 nights	2022
Magellan/LDSS3-C, 2 nights	2015

Awards

Future Investigators in NASA Earth, Space Sciences, and Technology Fellowship	2020-2021
Pierazzo International Student Travel Award	2017
University of Chicago Physical Science Teaching Prize (nominee x2)	2015, 2017

Service & Outreach

Journal Referee , Astronomical Journal Astronomy & Astrophysics Manuscripts refereed 1:1 commensurate with manuscripts published	2022-present
Time Allocation Committees Various committees, ongoing as requested	2022-present
Public Lectures Nerd Nite Los Angeles, “Why Aliens Will Look Like Us” Encyclopedia Comedia live panel show, episode “E is for Exoplanets”	2024 2024
Skype-a-Scientist Virtual Q&A with middle school science classroom (x3)	2021
Departmental Committees , University of Chicago Diversity, Equity, and Inclusion Advising/Mentorship subgroup Education and Outreach Teaching Physical Science Division Undergraduate Curriculum	2020 2019 2017 2017

Talks

★ ★ = colloquium / keynote speaker

Invited

California Institute of Technology Planetary Science Seminar	Pasadena, CA 2025
★ ★ University of Florida Astronomy Colloquium	Gainesville, FL 2025
Indiana University Astronomy Tea Talk	Bloomington, IN 2025
★ ★ LSST Discovery Alliance Data Science Fellowship Program	Pittsburgh, PA 2025
University of Hawai‘i / IfA Exoplanets Seminar	Mānoa, HI 2024
★ ★ Lowell Observatory Astronomy Colloquium	Flagstaff, AZ 2024
Princeton / IAS Extrasolar Planets Seminar	Princeton, NJ 2024
★ ★ NASA Jet Propulsion Lab Astronomy Colloquium	Pasadena, CA 2024
University of California, Los Angeles Planetary Science Seminar	Los Angeles, CA 2024
Center for Computational Astrophysics Lunch Seminar	New York, NY 2023
The Ohio State University Exoplanet Seminar	Columbus, OH 2021
Penn State University Astrophysics Lunch Talk	virtual 2020
University of California, Los Angeles Astronomy Lunch Talk	virtual 2020
DePaul University, “Life in the Universe” Seminar	Chicago, IL 2018

Contributed

Notre Dame University Exoplanets Group Meeting	South Bend, IN 2025
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Keck Science Meeting	Los Angeles, CA 2025
Center for Exoplanets and Habitable Worlds Seminar	State College, PA 2025
University of California, Santa Cruz Planetary Lunch Seminar	Santa Cruz, CA 2025
Know Thy Star 2 Conference	Pasadena, CA 2025
Extreme Solar Systems V Conference	Christchurch, NZ 2024
Columbia University Exoplanets Group Meeting	New York, NY 2023
Cool Worlds Lab Exomoons Conference	virtual 2020
Exoplanets III Conference	virtual 2020
Princeton / IAS Astronomy Seminar	virtual 2020
Lake Michigan Exoplanet Meeting	Chicago, IL 2019
Chicagoland Exoplanet Meeting	Chicago, IL 2017

Publications

SAO/NASA Astrophysics Data System list available at [gjgilbert.github.io/cv](https://github.com/gjgilbert/cv)

★ = student advisee

1st — 3rd Author

11. Orbital eccentricities suggest a gradual transition from giant planets to brown dwarfs

Gilbert, GJ · ★Van Zandt, JE · et al. *in review*

10. The transition from giant planets to brown dwarfs near the ice line from the stellar metallicity distribution

Giacalone, S · Howard, A · Gilbert, GJ · et al. *in review*

9. The occurrence of giant planets and brown dwarfs from RVs and astrometry

★Van Zandt, JE · Gilbert, GJ · et al. *in review*

8. The orbital eccentricity-radius relation for planets orbiting M dwarfs

★Sagear, S · Ballard, S · Gilbert, GJ · et al. *in review*

7. Planets larger than Neptune have elevated eccentricities

Gilbert, GJ · Petigura, EA · ★Entrican, PM 2025

6. The TESS-Keck Survey XV: Precise properties of 108 planets and their host stars

★MacDougall, MG · Petigura, EA · Gilbert, GJ · et al. 2023

5. Accurate and efficient photoeccentric transit modeling

★MacDougall, MG · Gilbert, GJ · Petigura, EA 2023

4. Implicit biases in transit models using stellar pseudo density

Gilbert, GJ · ★MacDougall, MG · Petigura, EA 2022

3. Accurate modeling of grazing transits using umbrella sampling

Gilbert, GJ 2022

2. Planetary period ratio sculpting near second-order mean motion resonances

Bailey, N · Gilbert, GJ · Fabrycky, DC 2021

1. An information theoretic framework for classifying exoplanetary system architectures

Gilbert, GJ · Fabrycky, DC 2020

Contributing Author

10. The orbital eccentricities of planets in the kinematic thin and thick galactic disks

★Sagear, S · et al. (including Gilbert, GJ) *in review*

9. A decade of transit-timing measurements confirm resonance in the K2-19 system	
★Entrican, PM · Petigura, EA · Petit, AC · Gilbert, GJ · Masuda, K	<i>in review</i>
8. TOI-6324b: An Earth-mass ultra-short-period planet transiting a nearby M dwarf	
Lee, RA · et al. (including Gilbert, GJ)	2025
7. The composition of rocky planets on close-in orbits tend to be Earth-like	
Brinkman, CL · et al. (including Gilbert, GJ)	2025
6. The TESS-Keck Survey XXIV: Outer giants may be more prevalent in the presence of inner small planets	
★Van Zandt, JE · et al. (including Gilbert, GJ)	2025
5. The prevalence of resonance among young, close-in planets	
Dai, F · et al. (including Gilbert, GJ)	2024
4. An Earth-sized planet on the verge of tidal disruption	
Dai, F · et al. (including Gilbert, GJ)	2024
3. The benchmark M dwarf eclipsing binary CM Draconis with TESS: spots, flares, and ultra-precise parameters	
Martin, DV · Sethi, R · Armitage, T · Gilbert, GJ · et al.	2024
2. The TESS-Keck Survey XIV: Two giant exoplanets from the distant giants survey	
★Van Zandt, JE · ★MacDougall, MG · Petigura, EA · Gilbert, GJ · et al.	2023
1. A search for water in the atmosphere of HAT-P-26b using LDSS-3C	
Stevenson, KB · Bean, JL · Seifahrt, A · Gilbert, GJ · et al.	2016